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Design of a charge-sensitive amplifier for MEMS piezoelectric hydrophone

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Abstract. Hydrophone is an important part of sonar system and convert acoustic signal into electrical signal. However, the electrical signal generated by the hydrophone is relatively weak and cannot be directly used for underwater detection and communication. Therefore, it is necessary to preprocess the generated electrical signal. In this paper, a switched capacitor charge-sensitive amplifier for MEMS piezoelectric hydrophone is designed. In the structure, two single-channel charge amplifiers are used for time-division multiplexing. The noise sampling and charge amplification are performed separately in the same half cycle to achieve the effect of alternating output, thereby increasing the sampling rate and solving the problem of discontinuous output of the switched capacitor circuit. In the circuit design, virtual switch and two-phase non-overlapping clock are adopted to reduce the error caused by charge injection and clock feedthrough effect of CMOS switch. What's more, the correlated double sampling technique is used to reduce the offset and low frequency noise of the circuit. The gain of the charge-sensitive amplifier designed in this study is 26 dB, the bandwidth is 0.8Hz~17.6kHz, the equivalent input noise power spectral density is 115.6 $nV/\sqrt{Hz}$ at 0.1Hz, and the charge sensitivity is 2 mV/pC. The fitting line between the input charge and the amplitude of the output curve is similar to the theory. The simulation results show that the charge-sensitive amplifier proposed in this study meets the design requirements of the MEMS piezoelectric hydrophone readout circuit.
Keywords. MEMS piezoelectric hydrophone; Charge-sensitive amplifier; Time division multiplexing; Correlated double sampling

1 Introduction

Sonar is an instrument used to detect and process underwater acoustic signals. It is widely used in underwater communication, topographic mapping, and target detection. It plays a vital role in national defense and civil fields [1,2]. Piezoelectric hydrophone fabricated by MEMS technology has the advantages of small size, compatibility with CMOS technology, easy mass production, high conversion efficiency and sensitivity, and no need for external power supply, it is an important part of underwater sonar system [3,4]. The principle of piezoelectric hydrophone is that the stress applied along a certain direction will polarize the piezoelectric material, resulting in charge [5]. Since the hydrophone is a high-impedance element and the generated signal is very weak, it cannot be directly used for underwater detection and communication, so a readout circuit is needed to preprocess the signal.

As the most common preamplifier in the readout circuit of piezoelectric hydrophone, the

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performance of charge-sensitive amplifier (CSA) directly affects the performance of the whole circuit [6]. Compared with the voltage amplifier, its advantages are that the amplification is not affected by the change of the cable, and the charge-voltage conversion accuracy and linearity are higher. Since the MEMS piezoelectric hydrophone works at low frequencies, the magnitude of the offset voltage and low-frequency noise from CSA directly related to the minimum amount of input charge that the chip can handle. Therefore, it is very important to study the CSA and improve its performance for the readout circuit.

The traditional CSA for piezoelectric sensors mostly use continuous-time circuits， and the performance of CSA is improved by improving the structure of operational amplifiers and high impedance circuits, without much suppression of noise and voltage offset [7,8]; the research of Alnasser E and D.P. Dobrev used active feedback circuits instead of high impedance circuits, but only reduced the output offset voltage [9,10].

To lift performance on offset voltage and low frequency noise, a switched capacitor (SC) charge-sensitive amplifier for MEMS piezoelectric hydrophone is designed in this paper. The CSA applies correlated double sampling (CDS) structure to reduce the magnitude of the offset voltage and low frequency noise. At the same time, two single-channel charge amplifiers are used for time-division multiplexing to improve the sampling rate and solve the problem of discontinuous output of switched capacitor circuits. The simulation results of the proposed structure show that the equivalent input noise power spectral density (PSD) at 0.1 Hz is 115.6 $nV/\sqrt{Hz}$, and it has good linearity.

2. CSA principle and circuit implementation

2.1 Interface Circuit of CSA

Before improving the traditional structure, this section will analyze the basic principle of the CSA and derive the main formulas and indicators.

Piezoelectric hydrophone can be equivalent to a voltage source in series with a large capacitor or a charge source in parallel with a large capacitor, so it can be regarded as a high impedance signal source with weak output signal. Therefore, the preamplifier circuit needs to have a high input impedance [11]. The circuit has two main functions: it can not only amplify the signal, but also transform the sizable output impedance of the sensor itself into a small sized impedance. In addition, it can also prevent the rapid leakage of charge from causing measurement errors.

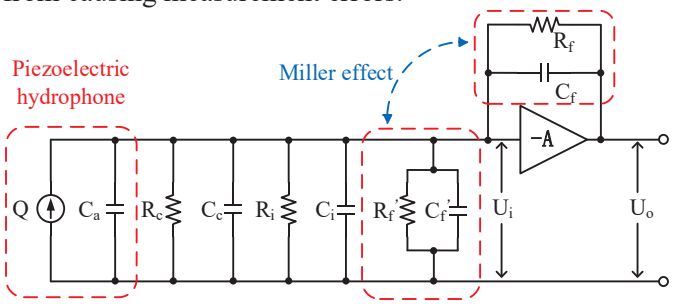


Figure 1. Schematic diagram of the interface circuit for Piezoelectric hydrophone

As shown in Figure 1, the charge-sensitive amplifier is the most common preamplifier circuit for MEMS piezoelectric hydrophones, where C_f and R_f are the feedback capacitance and resistance, C_i and R_i are the input capacitance and resistance of the operational amplifier, C_a is the equivalent capacitance of the piezoelectric hydrophone. The insulation resistance R_a of the hydrophone is between $10^9 \sim 10^{14}$ Ω, which can be considered as open circuit relative to other parallel resistors, C_c and R_c are cable capacitance and cable resistance.

According to the “Miller effect”, the equivalent capacitance of the feedback capacitor converted to the input end of the operational amplifier is:

$$C_f' = C_f(1+A) \tag{1}$$

and the equivalent resistance of the feedback resistor to the input end of the amplifier is:

$$R_f' = R_f / (1 + A) \tag{2}$$

When the gain of the amplifier is large enough, the voltage on the output end of the charge amplifier can be denoted as the following formula:

$$U_o = \frac{-jwAQ}{(\frac{1}{R_c} + \frac{1}{R_i} + \frac{1+A}{R_f}) + jw[C_c + C_i + C_a + (1+A)C_f]} \approx \frac{-jwAQ}{(1+A)(1/R_f + jwC_f)} \approx \frac{-Q}{C_f} \tag{3}$$

The establishment of the above formula is based on the fact that the gain A is large and the $1/R_f \ll |wC_f|$ holds. When the frequency is very low, there will be $1/R_f = |wC_f|$, in the other word:

$$f_L = \frac{1}{2\pi R_f C_f} \tag{4}$$

and the corresponding output voltage is:

$$U_o = \frac{-1}{\sqrt{2}} \frac{Q}{C_f} \tag{5}$$

Therefore, in addition to the upper cut-off frequency, the CSA for piezoelectric hydrophones has a lower cut-off frequency at low frequencies. Usually, the lower the frequency, the greater the range of the circuit can be measured. When the frequency is close to 0, it is called "Quasi-static charge amplifier".

Formula (3) shows that compared with the voltage amplifier, the output end of the CSA is not sensitive to the change of the cable capacitance C_c, so the replacement of the cable has little effect on the sensitivity of the sensor. In addition, the voltage on the CSA's output end and the amount of charge generated by the sensor shows a proportional relationship, so CSA is more suitable for the measurement of piezoelectric sensors.

2.2 Three-stage operational amplifier

The power supply voltage of this design is 5V, and the piezoelectric coefficients of piezoelectric hydrophones of different materials are different. Therefore, in order to avoid output voltage exceeding the voltage output range, sometimes it is necessary to use a feedback capacitor close to NanoFarad level to set the appropriate magnification, which puts forward requirements for the load driving ability of the operational amplifier in CSA.

Compared with the traditional CSA, this paper uses CMOS process to integrate the whole readout circuit. However, a two-stage CMOS operational amplifier requires large current and area to achieve phase compensation when driving a large capacitive load. In contrast, the three-stage operational amplifier can ensure stability with lower power consumption and smaller chip area [12]. Figure 2 shows the Dual-miller Compensation with Embedded Current Amplifier (DCECA) in this paper.

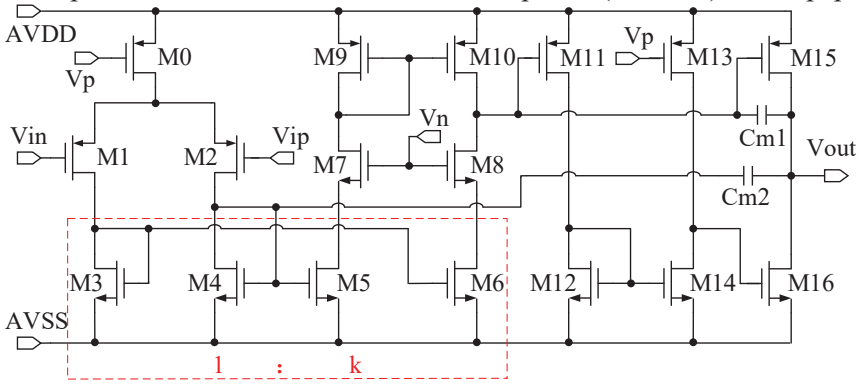


Figure 2. Structure of the three-stage operational amplifier used in this paper

The input pair M1 and M2 of DCECA use PMOS differential pairs with less 1/f noise. The current amplifier composed of M3 ~ M10 is embedded between the first stage and the second stage amplifier to

realize the current multiplication function and reduce the area of the compensation capacitor C_{m2}. The capacitor C_{m1} and capacitor C_{m2} constitute two compensation networks. The C_{m1} branch can ensure the stability of the amplifier when it is connected to a small capacitive load, and the C_{m2} branch is used to compensate for a larger capacitive load.

The magnification of DCEAC can be expressed by the following formula

$$A_V \approx \frac{A_{dc}(1+s\frac{C_{m2}}{2g_{mf}})(1-s\frac{C_{m1}}{g_{m2}g_{m3}R_2}+s^2\frac{C_{m1}C_2}{g_{m2}g_{m3}})}{(1+s|C_m g_{m2}g_{m3}R_1R_2R_3|)(1+s\frac{C_{m1}C_{m2}}{g_{mf}C_m}+s\frac{C_{m1}C_L}{g_{m2}g_{m3}R_2C_m}+s^2\frac{C_{m1}C_{m2}C_L}{g_{m2}g_{m3}g_{mf}R_2C_m}+s^3\frac{C_{m1}C_{m2}C_LC_2}{g_{m2}g_{m3}g_{mf}C_m})} \quad (6)$$

in the above formula, $C_m = C_{m1} + kC_{m2}$, R_i and C_i are the output loads of each stage, g_{mf} is the transconductance of the current amplifier, C_L is the load capacitance, and the low frequency gain $A_{dc} = kg_{m1}g_{m2}g_{m3}R_1R_2R_3$.

Formula (6) shows that after compensation, the factor containing C_L are located at the higher-order poles of the operational amplifier, so the C_L has little effect on the phase margin. Therefore, the operational amplifier with this structure can drive a wide range of captance from PicoFarad to NanoFarad level with a small area of compensation capacitor.

2.3 Circuit implementation of CSA

Piezoelectric hydrophones mainly work in the low frequency band, so it is greatly affected by 1/f noise and offset voltage. Therefore, the low frequency characteristics can be greatly improved by suppressing them through the CDS technique.

The CSA circuit and timing diagram designed in this paper are shown in Figure 3. It can be seen from the A part of this section that the magnification can be changed by controlling the value of C_f. R_f is a high impedance, which is responsible for determining the cut-off frequency of the amplifier, stabilizing the DC operating point, and venting the charge on the C_f. In addition, a larger R_f can reduce the phase displacement between the input and output voltage. The resistance and capacitance of the positive input port of the amplifier are used to guarantee the symmetry of the CSA. The right half of the circuit is a switched capacitor buffer using correlated double sampling technique, it is used to drive and match the post-stage circuit.

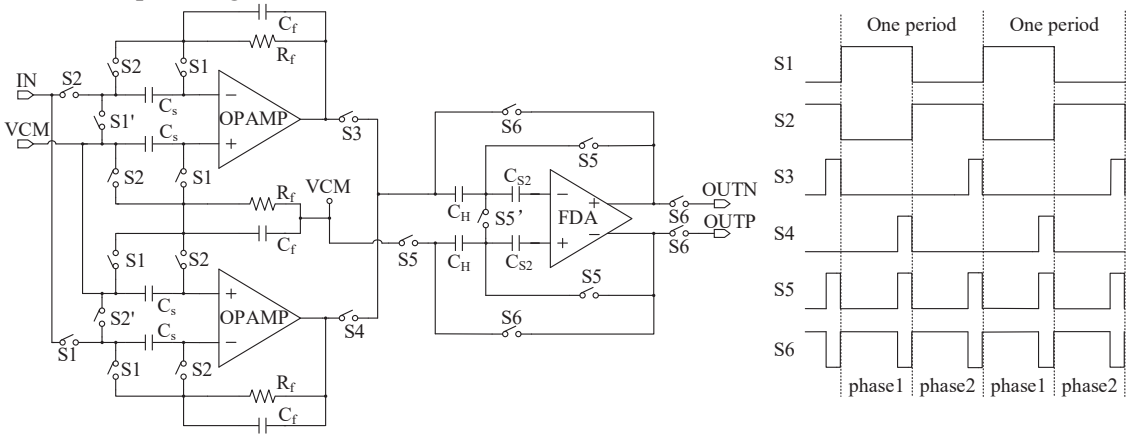


Figure 3. Structure of the proposed CSA and the switch signal

The CSA achieves correlated double sampling by switching the conduction of switches S1 and S2, and reduces the 1/f noise and offset voltage by changing the conduction order and using a two-phase non-overlapping clock. The switch S1'and S2'are disconnected slightly later than S1 and S2, respectively, which can diminish the influence of charge injection effect. The switch S3 and S4 cooperate with the switch S1 and S2 for gating, so that the two single-channel charge amplifiers perform noise

sampling and charge amplification respectively in the same half-cycle, and there is always a signal alternately output to the buffer in each stage, which improves the sampling rate and improves the problem of discontinuous output of traditional switched capacitor circuits. The sampling stage of the buffer overlaps with the charge amplification stage of the main circuit, and the output voltage of S3 or S4 is transmitted to OUTN and OUTP to drive the subsequent circuit. The principle of the buffer to eliminate 1/f noise and offset voltage is similar to the main part of CSA. The following is only the theoretical analysis of the main circuit of CSA:

In one cycle, the circuit is divided into two working stages: noise sampling and charge amplification. Taking half-circuit as an example, the derivation of charge amplification and noise suppression is carried out.

During the noise sampling phase stage, the switch S1 is turned on and S2 is turned off. The left port of the C_S is connected to the VCM to sample the 1/f noise and offset voltage. The feedback resistance R_f releases the charge on the capacitor C_f. Within this stage, the switch S3 is turned off, and the CSA of another channel outputs voltage through the switch S4.

During the charge amplification phase stage, the switch S2 is turned on, S1 is turned off, and the left port of the C_S is connected to the input port. When the switching period is short enough and the noise correlation between the two stages is high enough, the low-frequency noise and voltage offset can be eliminated from the left side of the feedback capacitor. Within this stage, the input charge is transferred to C_f and output through the switch S3.

Deriving the above process, during the noise sampling stage, the main circuit of the CSA is shown in Figure 4 (a), where V_n is the sum of low frequency noise and offset voltage. During $(n-1)T \sim (n-1/2)T$:

$$V_-[(n-1/2)T] = V_+[(n-1/2)T] = V_{CM} \tag{7}$$

$$V_{CS}[(n-1/2)T] = V_n[(n-1/2)T] \tag{8}$$

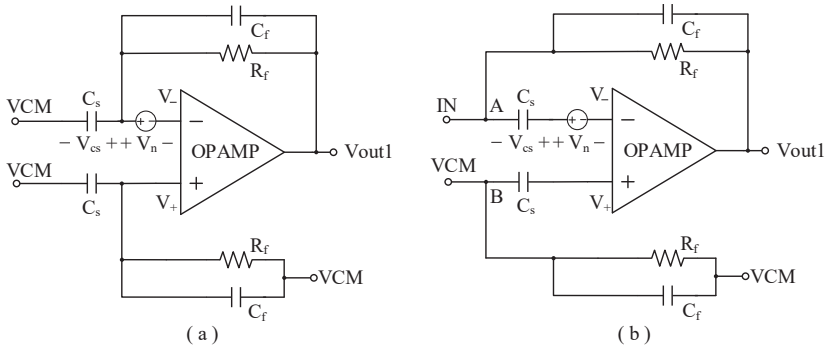


Figure 4. Simplified schematic of single-channel CSA:
(a) Noise sampling phase stage (b) Charge amplification phase stage

The charge amplification stage circuit is shown in Figure 4 (b). The switching frequency is high enough, so the low-frequency noise in the two stages has a strong correlation and the value of it is basically unchanged, what's more, the offset voltage is a fixed value, so V_n can be regarded as unchanged, then within $(n-1/2)T \sim nT$:

$$V_n[(n-1/2)T] = V_n[nT] \tag{9}$$

$$V_-[nT] = V_+[nT] = V_{CM} \tag{10}$$

$$V_A[nT] = V_-[nT] + V_n[nT] - V_{CS}[(n-1/2)T] \tag{11}$$

The above formula can deduce:

$$V_A[nT] = V_{CM} \tag{12}$$

$$NTF(z) = \frac{V_A(z)}{V_n(z)} = (1 - z^{-\frac{1}{2}}) \tag{13}$$

NTF (z) in formula (13) represents the noise transfer function from V_n to point A, which is characterized by high-pass characteristics. It can be seen in both the time domain and frequency domain that this design can filter out the offset voltage and low-frequency noise at point A.

When the charge transfer is completed, we can get:

$$\Delta V_{OUT}[nT] = \frac{\Delta Q}{C_f} \tag{14}$$

From the above derivation, it can be seen that after two stages of processing, the low-frequency noise and offset voltage have been eliminated from the output port, and CSA can more accurately amplify the weak charge generated by MEMS piezoelectric hydrophone.

2.4 Non-ideal factor analysis and circuit optimization

The main non-ideal effects that affect the performance of the readout circuit include low-frequency 1/f noise, device mismatch, limited gain of the operational amplifier, charge injection effect, and clock feedthrough effect. This section optimizes the main non-ideal factors to improve circuit performance.

2.4.1 The non-ideal effect introduced by operational amplifier. The three-stage operational amplifier structure used in this design is shown in Figure 2. The noise, offset voltage and device mismatch of the operational amplifier will introduce errors. In the circuit, noise primarily consists of thermal noise and 1/f noise. The noise of a MOSFET can be expressed as:

$$\overline{V_{n,tot}^2} = \frac{4kT\gamma}{g_m} + \frac{K}{C_{ox}(WL)}\frac{1}{f} \tag{15}$$

When calculating the equivalent input noise, the noise from second and third stages of the amplifier mentioned in Figure 2 can be ignored. The noise of the first stage is mainly contributed by M1~M6, M9 and M10. Equating the noise of each MOSFET to the input, the PSD of input noise of the amplifier is:

$$\begin{aligned}\overline{V_{n,in}^2} = {} & \frac{8kT\gamma}{g_{m1,2}}\left(1+\frac{g_{m3,4}}{g_{m1,2}}+\frac{g_{m5,6}}{g_{m1,2}}+\frac{g_{m9,10}}{g_{m1,2}}\right) \\ & +\frac{2K_P}{C_{ox}f}\left[\frac{1}{(WL)_{1,2}}+\frac{g_{m9,10}^2}{g_{m1,2}^2}\frac{1}{(WL)_{9,10}}\right]+\frac{2K_N}{C_{ox}f}\frac{1}{g_{m1,2}^2}\left[\frac{g_{m3,4}^2}{(WL)_{5,6}}+\frac{g_{m5,6}^2}{(WL)_{5,6}}\right]\end{aligned} \tag{16}$$

Formula (16) shows that the thermal noise and 1/f noise generated by the operational amplifier can be reduced by increasing $g_{m1,2}$ and reducing $g_{m3,4}$, $g_{m5,6}$ and $g_{m9,10}$.

2.4.2 The non-ideal effect introduced by the switch. Formula (13) shows that the value of the C_S has no effect on the elimination of 1/f noise and offset voltage, but in fact, the larger the C_S, the better the suppression result of the circuit on 1/f noise and offset voltage, which is due to the fact that the switch used in the circuit is not an ideal device.

The switch composed of MOSFETs has non-ideal effects such as clock feedthrough and channel charge injection, which affects the accuracy of the switched capacitor circuit. Two non-ideal effects are shown in Figure 5.

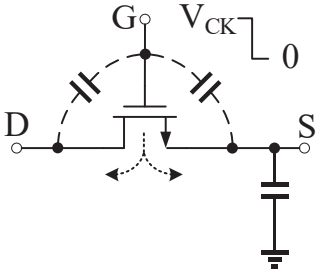


Figure 5. Non-ideal effects in MOS switches

The clock feed through effect refers to the error generated by the parasitic capacitance at the source

and the drain port when the gate voltage of the MOSFET jumps. The value is:

$$\Delta V = V_{EN}\frac{WC_{ov}}{WC_{ov}+C_H} \tag{17}$$

V_{EN} is the amplitude of the clock, and C_{OV} is the overlapping capacitance per unit width. Formula (17) shows that the deviation triggered by the clock feedthrough effect is a fixed offset.

When a MOSFET is shut down, the charge in the inversion layer channel will leak from the source and the drain port, causing the channel charge injection effect. The sum of the outflow charges is:

$$Q = WLC_{ox}\left(V_{EN}-VIN-V_{THN}\right) \tag{18}$$

C_{ox} is the gate oxide layer capacitance per unit area. The charge injection effect causes gain error, nonlinear error and DC offset, thus affect the accuracy of the circuit. When the NMOS switch is used, the outflow charge causes a negative pressure drop at the sampling capacitor; when a PMOS switch is used, the outgoing charge causes a positive voltage drop at the sampling capacitor, and the steps of these two voltage drops are in opposite directions. Therefore, CMOS switches with dummy can be used to improve the channel charge injection effect and clock feedthrough effect.

Figure 6 is the switch used in this paper. M1 and M2 constitute a CMOS switch, the switch has a larger input dynamic range than a single MOSFET. M3-M6 are dummy MOSFETs, which are used to offset the effects of partial clock feedthrough and charge injection. EN and ENB are a pair of complementary clocks to reduce the distortion when the switch is turned off.

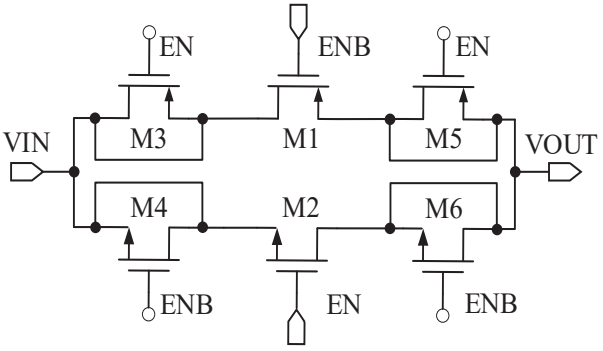


Figure 6. Switch structure of the proposed CSA

The impedance of CMOS switch is

$$R_{on} = R_{onN} \,||\, R_{onP} = \frac{1}{\mu_n C_{ox}\frac{W}{L}\left(V_{EN}-VIN-V_{THN}\right)} \,||\, \frac{1}{\mu_n C_{ox}\frac{W}{L}\left(VIN-V_{THP}\right)} \tag{19}$$

Formula (18) shows that the deviation triggered by charge injection can be reduced by reducing the area of the switch. However, if the width-to-length ratio is too small, the resistance of the switch will increase, which will not only increase the thermal noise of the switch, but also lead to a decrease in the switching speed when the sampling capacitance is constant. Therefore, the design of the switch should be a compromise between speed and accuracy.

3. Result and discussion

The CSA proposed in this paper is simulated based on DB Hitek 0.18μm CMOS process. The simulation results of the operational amplifier are shown in Figure 7. The gain is 89dB, when k=4 and the load capacitance is 2nF, only 2.2pF Miller compensation capacitor is needed to achieve a phase margin of 77 °, which saves the chip area. The equivalent input noise power spectral density is 29.52 $nV/\sqrt{Hz}$ at 1MHz.

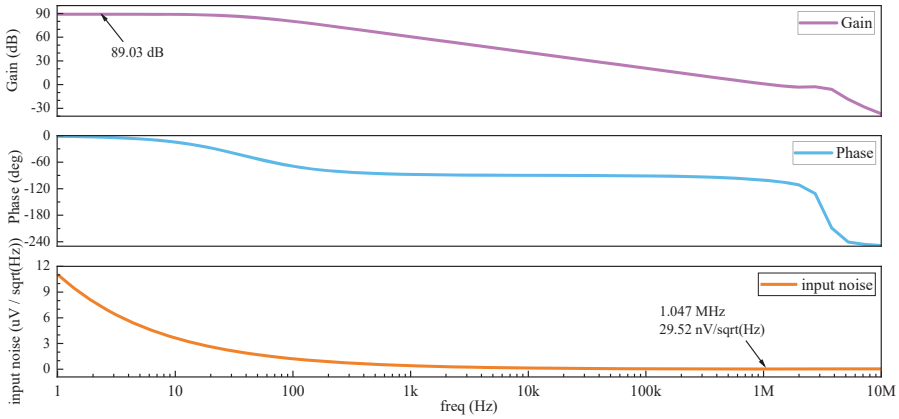


Figure 7. Simulation results of three-stage operational amplifier

The gain curve of CSA obtained by PSS and PAC simulation is shown in Figure 8. When the feedback capacitance C_f is 500 pF and the capacitance C_a of the hydrophone is 10nF, the gain of the charge amplifier is 26.02 dB, and the passband is from 0.8 Hz to 17.6 kHz. By adjusting the size of the feedback capacitance C_f and feedback resistance R_f, the magnification and bandwidth can be controlled to match the piezoelectric hydrophones of different materials.

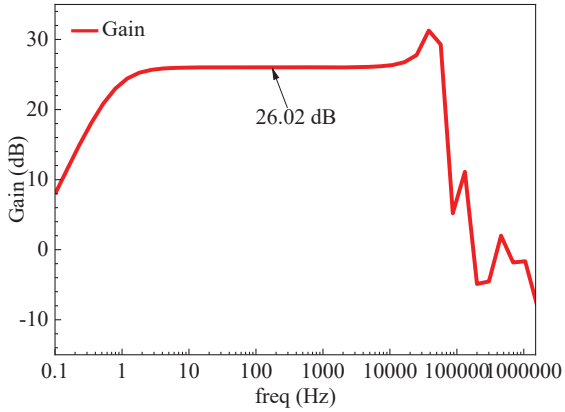


Figure 8. Simulation results of CSA gain curve

When the input frequency is 2kHz sine wave, the output voltage of CSA is shown in Figure 9 red curve, and the blue curve is an ideal sine wave curve. It can be identified from the comparison diagram that the CSA designed in this article can better amplify the electrical signals generated by the piezoelectric hydrophone.

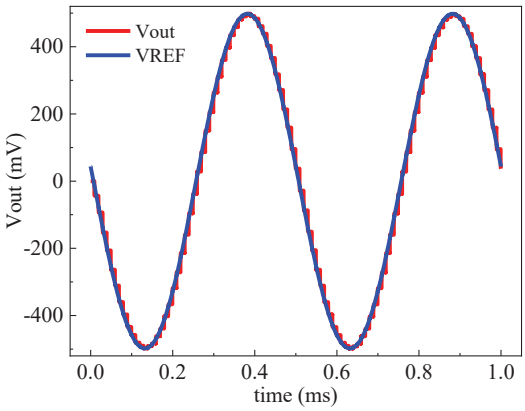


Figure 9. Comparison of the output voltage and the ideal sine wave

The PSD of equivalent input noise curve is shown in Figure 10. The PSD of the equivalent input noise at 0.1 Hz is 115.6 $nV/\sqrt{Hz}$. The noise curve is flat in the passband, indicating that the CSA eliminates low-frequency noise, and the remaining part is almost all thermal noise.

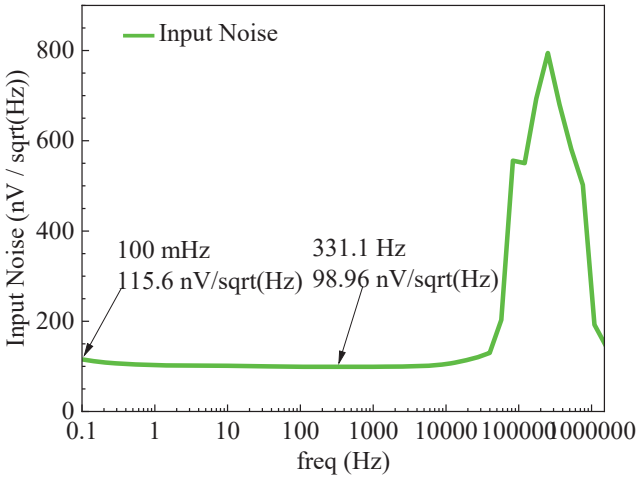


Figure 10. Equivalent input noise simulation of CSA

The charge amplification factor is simulated by using a pulsed current source to instead of a charge source. The input charge was fitted to the output amplitude using Origin software and the fitted straight line is shown in Figure 11. When the charge sensitivity is set to 2 mV/pC, the input charge and the output voltage maintain a good linear relationship, and the slope of the fitting line is approximately equal to the charge sensitivity, the charge-sensitive amplifier designed in this paper has good linearity.

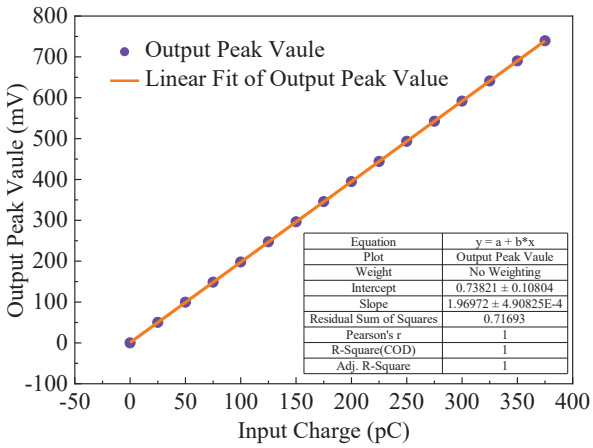


Figure 11. The fitting diagram of the relationship between the input charge and the output curve amplitude

4. Conclusion

In this research, a SC charge-sensitive amplifier for MEMS piezoelectric hydrophone is designed. The amplifier is based on CDS technology, which suppresses low frequency noise and voltage offset, and reduces channel charge injection effect and clock feedthrough effect. The time-division multiplexing structure is adopted in the circuit, which improves the sampling rate and solves the problem of discontinuous output of the traditional switched capacitor circuit.

The gain of the CSA put forward in this study is 26 dB, the bandwidth is 0.8Hz-17.6kHz, the PSD of the equivalent input noise is 115.6 $nV/\sqrt{Hz}$ at 0.1Hz, and the charge sensitivity is 2 mV/pC. The fitting line between the input charge and the amplitude of the output curve is similar to the theory. It is proved that the charge-sensitive amplifier proposed in this paper has low noise and good linearity, which

can meet the performance requirements of MEMS piezoelectric hydrophone readout circuit.

Acknowledgments
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